

PROBLEM AND OBJECTIVES

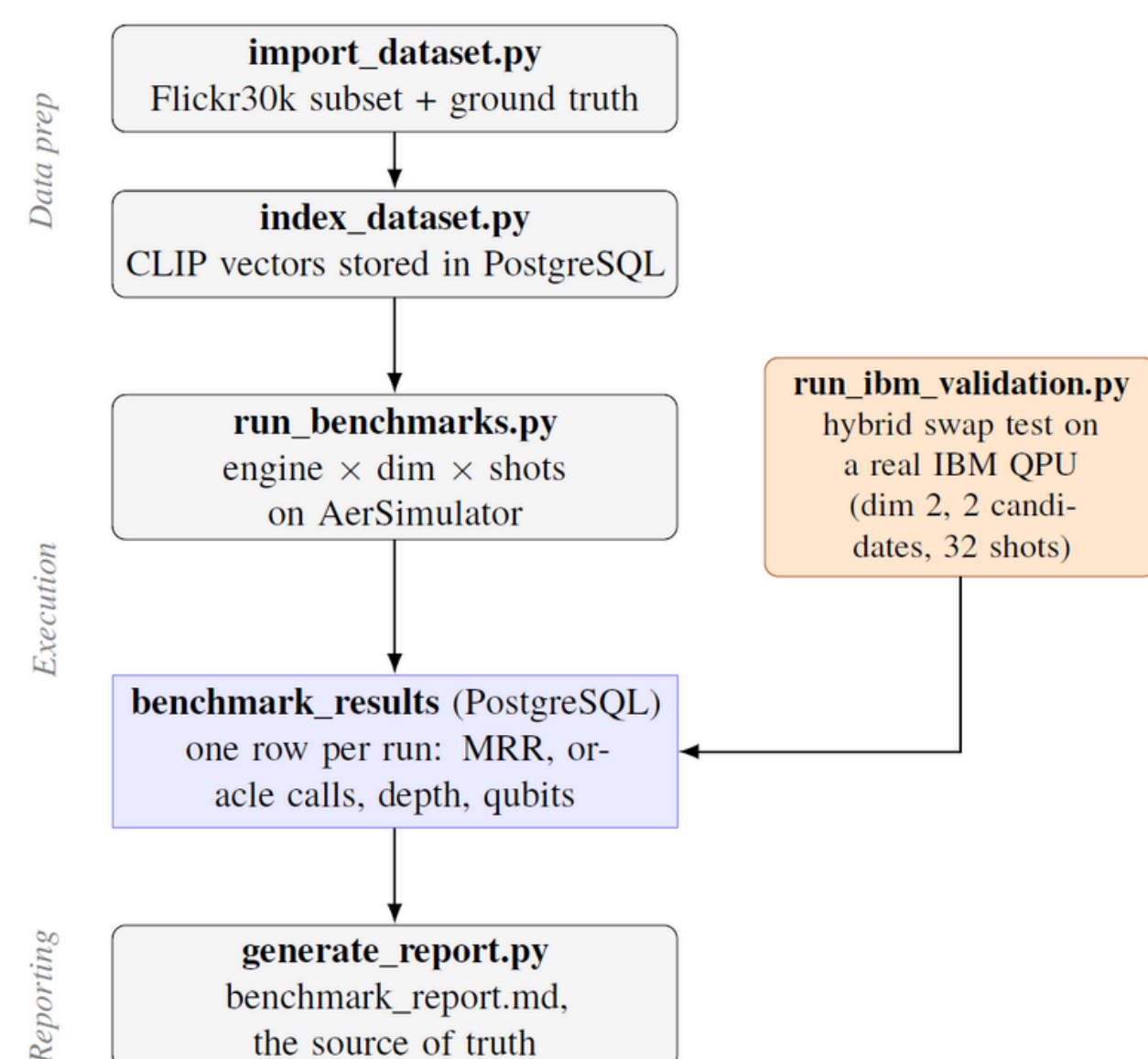
Information retrieval systems rely on vector similarity search to find the closest match to a query across multiple items. However, classical methods are limited to $O(N)$ cost, which become infeasible as datasets grow. This project addresses whether quantum computing can offer improvements over classical methods.

Objectives: to research and answer the following:

1. **Accuracy:** Does the swap-test reproduce classical cosine ranking, and at what shot budget?
2. **Scaling:** Does our Grover implementation match $\frac{\pi\sqrt{N}}{4}$ oracle-call curve on real circuits?
3. **Practicality:** Can a quantum engine beat a production HNSW baseline?
4. **Suitability:** Is vector retrieval a good problem class for quantum computing?

Quantum computing *promises* speedups, but we test it in practice.

METHODOLOGY AND DESIGN



4 scripts pipeline: Import data set, pair images to ground truth queries → vector conversion-CLIP embeddings → seven engines → PostgreSQL benchmark rows → FastAPI + React results.

Engine	Type	Complexity
brute_force_cosine	Classical	$O(N)$
faiss_flat_l2	Classical	$O(N)$
faiss_hnsw_l2	Classical	$O(\log N)$
qiskit_swap_test	Quantum	$O(N)$
qiskit_grover	Quantum	$O(\sqrt{N})$
qiskit_grover_quantum_prep	Quantum	$O(\sqrt{N})$
hybrid_hnsw_swap_test	Hybrid	$O(\log N + M)$
hybrid_hnsw_swap_test_ibm	IBM QPU	Validation only

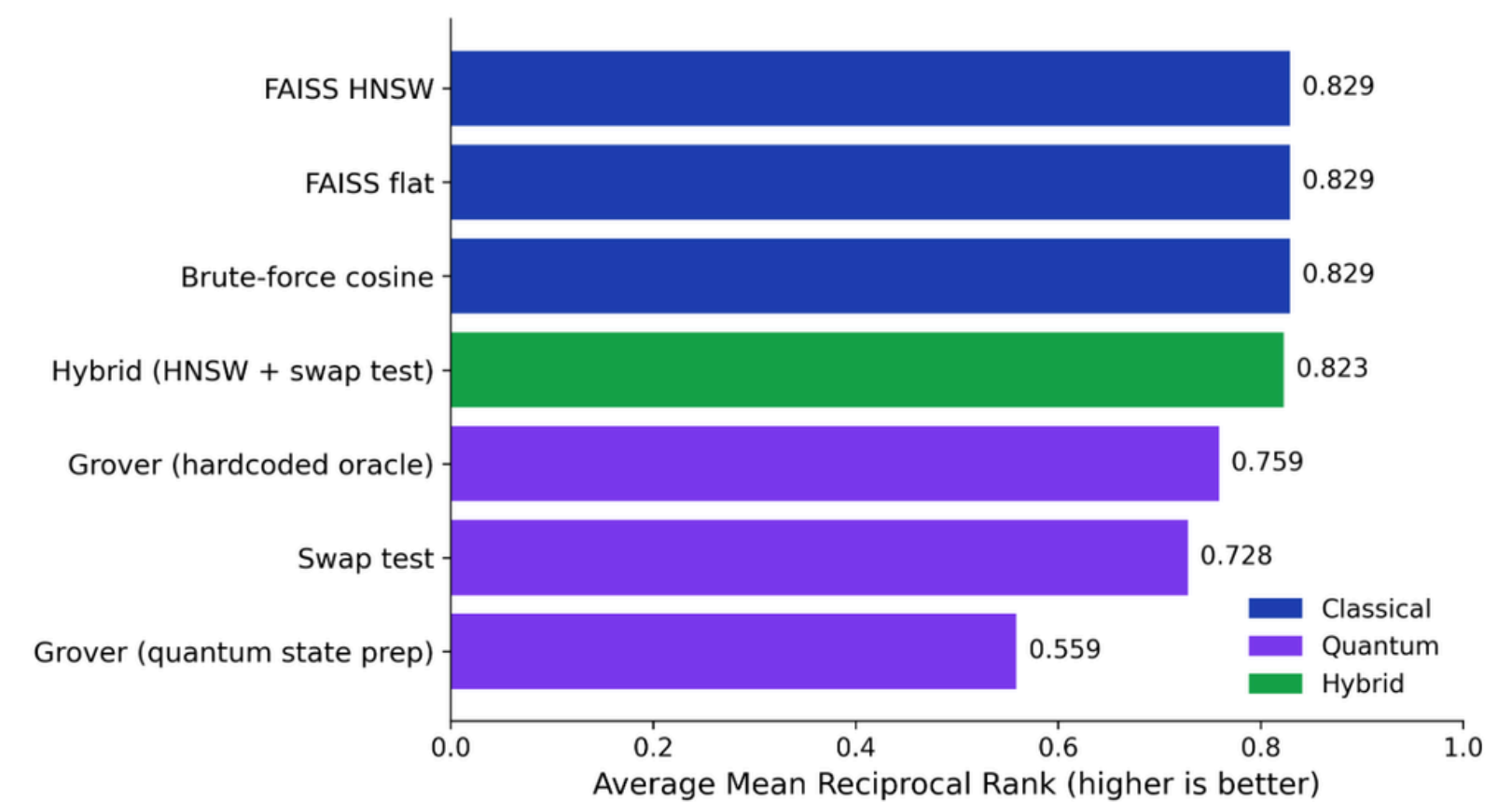
Main design components are the **eight engines** with different methods and definitions.

IMPLEMENTATION & RESULTS

Technology stack: Python, FastAPI, React, FAISS, PostgreSQL/pgvector, Qiskit Aer + IBM Quantum

Main **evaluation metrics** (Cross-engine):

- Mean Reciprocal Rank (MRR)- Average of $1 / \text{rank}$ of first relevant result across queries. Higher is better.
- Oracle calls- Number of similarity comparisons per query. N for classical and swap test, $\frac{\pi\sqrt{N}}{4}$ for Grover.



Classic methods achieve the highest accuracy: MRR= 0.829. Hybrid comes close at 0.823. Quantum approaches perform the worst at 0.559, indicating that classical and hybrid methods currently outperform quantum pipelines (Average MRR across all dimensions).

Engine	Oracle calls	Engine	Oracle calls
BF Cosine	20	Grover (hardcoded oracle)	4
FAISS flat	20	Grover (quantum prep)	4
FAISS HNSW	5	Hybrid (HNSW+swap test)	10
Swap test	20		

Comparing number of oracle calls per query for $N=20$ (dataset), classical methods needed 20 calls, while FAISS HNSW reduces it to 5. Quantum methods are the most efficient- 4 oracle calls each. The hybrid HNSW+swap test approach requires 10.

Grover's algorithm in our implementation **matched the theoretical formula** $\frac{\pi}{4} \sqrt{N}$, using 4 oracle calls for the padded dataset size of $N'=32$.

IBM hardware test achieved a low MRR of 0.125 due to truncation (2D vectors), low candidates (2), and noise. This validated the quantum pipeline but was not designed to demonstrate scaling or performance advantages.

CONCLUSION

Accuracy: Swap test matched classical cosine ranking, but required higher shot budgets at lower dimensions.

Scaling: Our Grover implementation exactly matched the theoretical oracle-call curve (4 calls for $N'=32$).

Practicality: No quantum engine outperformed the classical HNSW baseline on this dataset.

Suitability: Vector retrieval is not a natural fit for quantum computing because it is data-bound, not compute-bound.

Classical wins: Quantum methods matched theory but couldn't outperform in practice due to data-loading costs.

References

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2. Johnson, J., Douze, M., & Jegou, H. (2019). Billion-scale similarity search with GPUs. IEEE Transactions on Big Data, 7(3), 1–1. <https://doi.org/10.1109/tbdata.2019.2921572>
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